

LD6 Multilevel Density Research-Gate Specification

Phase-robust coarse/fine profiling, evidence policy, and the decision to retain the single-level architecture

mdstats development specification

2026-07-20

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Status and scope

This specification governs architecture gate **LD6** for `mdstats`. It begins from `mdstats 0.19.51a0`, where dense, single-level block-sparse, automatic backend selection, periodic sparse rendering, exact scientific planning, and bounded sparse optimization are complete for atomic, framework-vertex, and framework-edge density fields.

Implementation status: completed in `mdstats 0.19.52a0`.

LD6 is an evidence gate. It does not add a production multilevel field. It adds a bounded, deterministic research profiler that asks whether a dyadic coarse/fine hierarchy offers enough additional benefit over the completed dense plus single-level block-sparse architecture to justify a separate specification for transfer, conservation, HDR integration, and crack-free coarse/fine contouring.

The implemented evidence policy concludes:

```
retain_single_level
```

Therefore no production multilevel AMR implementation is authorized by this stage.

Scientific and architectural question

The completed architecture already chooses between:

1. a dense periodic logical-node field for broad support; and
2. a single-level block-sparse field for localized support.

A multilevel hierarchy is justified only if representative fields remain poorly served by both choices and if a phase-robust coarse/fine surrogate provides a substantial additional storage reduction while preserving field and HDR semantics.

The patch-hierarchy motivation follows block-structured local adaptive refinement (Berger and Colella, 1989). The periodic phase sweep, HDR-driven refinement rule, conservative averaging, evidence thresholds, and decision policy are project-specific `mdstats` definitions. The profiler is deliberately optimistic: passing it is necessary but not sufficient for a future production hierarchy.

Non-objectives

LD6 does not implement:

- multilevel field storage;
- prolongation or restriction operators used in production;
- coarse/fine Gaussian convolution;
- multilevel HDR integration;
- adaptive marching cubes or dual contouring;
- coarse/fine crack stitching;
- a new backend identifier;
- changes to the LD4 selector;
- changes to density values, meshes, or scene schemas.

No output of the profiler may be passed to a renderer as a scientific field.

Public research API

```
@dataclass(frozen=True)
class MultilevelResearchOptions:
    coarsening_factors: tuple[int, ...] = (2, 4)
    fine_mass_fractions: tuple[float, ...] = (0.90, 0.95, 0.99)
    block_shapes: tuple[tuple[int, int, int], ...] = (
        (4, 4, 4),
        (8, 8, 8),
        (16, 16, 16),
    )
    hdr_fractions: tuple[float, ...] = (0.50, 0.80, 0.95)
    max_relative_ll_error: float = 2.0e-3
    max_relative_linf_error: float = 1.0e-2
    max_relative_hdr_threshold_error: float = 1.0e-2
    max_hdr_mass_fraction_error: float = 1.0e-3
    minimum_incremental_storage_reduction: float = 2.0
    localized_active_fraction: float = 0.20
    broad_active_fraction: float = 0.50
    minimum_adoption_cases: int = 2
    max_profile_nodes: int = 4_000_000
    max_phase_evaluations: int = 2_000
    max_workspace_bytes: int = 512_000_000
```

Primary entry points are:

```
profile_multilevel_field(field, *, options=None)
decide_multilevel_research(profiles, *, options=None)
```

The profiler accepts any realized scalar field satisfying `ScalarField3D` and `PeriodicNodeFieldAccess`. All records are immutable, schema-versioned, canonically JSON-serializable, and independent of renderer state.

Baseline single-level profiling

For a logical grid with $N = N_1 N_2 N_3$ nodes, let N_+ be the positive-node count. The active fraction is

$$f_+ = \frac{N_+}{N}.$$

For the realized backend with N_s stored scalar slots, the stored fraction is

$$f_s = \frac{N_s}{N}.$$

Support is classified as:

```
f+ <= 0.20  -> localized
f+ >= 0.50  -> broad
otherwise   -> intermediate
```

The profiler recomputes exact single-level block-packing plans for 4^3 , 8^3 , and 16^3 blocks. This prevents block padding in one user-selected block size from being misidentified as evidence for a multilevel hierarchy.

A field is considered already served by the completed architecture when:

- broad support uses the dense backend; or
- localized support uses block-sparse storage and either the realized block shape or one profiled single-level block shape stores no more than 20% of the logical grid.

This is an architecture-sufficiency classification, not a claim that one block shape is universally fastest.

Dyadic coarse/fine research surrogate

Eligibility

A coarsening factor $r \in \{2, 4\}$ is profiled only when every logical-grid dimension is divisible by r . A production implementation for nondivisible hierarchies would require a separate boundary and ownership specification.

Fine-region selection

For requested fine mass fraction q_f , let c_{q_f} be the field's existing HDR threshold. Every positive node with

$$\rho_i \geq c_{q_f}$$

marks its coarse bin as fine-owned. A fine-owned bin retains all r^3 logical-node slots exactly, including implicit zeros. This is conservative with respect to local fine support and intentionally overestimates fine storage near threshold boundaries.

Coarse-region restriction

A non-fine bin B is represented by one piecewise-constant value

$$\bar{\rho}_B = \frac{1}{r^3} \sum_{i \in B} \rho_i.$$

The reconstructed fine-grid mass in the bin is exactly

$$r^3 \bar{\rho}_B \Delta V = \sum_{i \in B} \rho_i \Delta V.$$

Thus the surrogate preserves total measure to floating-point tolerance. Fine-owned bins are exact; only coarse-owned bins contribute approximation error.

Periodic coarse-grid phase sweep

A dyadic hierarchy anchored at one logical origin is not translation-neutral. For each factor r , LD6 evaluates every periodic coarse-grid phase

$$\mathbf{p} \in \{0, \dots, r-1\}^3.$$

There are r^3 phases. A candidate passes only when **every** phase satisfies the scientific tolerances. Storage benefit is evaluated using the worst phase. This avoids approving a hierarchy whose apparent advantage depends on an arbitrary coarse-grid origin.

Error and HDR criteria

For the reconstructed fine-grid surrogate $\tilde{\rho}$, LD6 requires

$$\frac{\|\tilde{\rho} - \rho\|_1}{\|\rho\|_1} \leq 2 \times 10^{-3},$$

$$\frac{\|\tilde{\rho} - \rho\|_\infty}{\max \rho} \leq 10^{-2}.$$

For HDR fractions $q \in \{0.50, 0.80, 0.95\}$,

$$\frac{|\tilde{c}_q - c_q|}{\max \rho} \leq 10^{-2},$$

and the achieved-mass-fraction difference is at most 10^{-3} .

Total-measure error is bounded by

$$|\tilde{M} - M| \leq 5 \times 10^{-13} \max(1, M).$$

HDR calculations account exactly for the multiplicity r^3 of one coarse value. Implicit zero nodes do not affect positive thresholds but are included in the L^1 and L^∞ reconstruction error.

Optimistic storage estimate

For one phase, let N_f be the number of fine-owned bins and N_c the number of positive coarse-owned bins. The optimistic value count is

$$N_{\text{ML}} = r^3 N_f + N_c.$$

The estimated bytes include float64 values and one three-int64 region index per fine or coarse region:

$$B_{\text{ML}} = 8N_{\text{ML}} + 24(N_f + N_c).$$

This estimate excludes transfer stencils, refinement masks, tree links, ghost zones, coarse/fine interpolation state, and adaptive-mesh contouring state. It is therefore a lower bound. Failure to show a strong gain under this optimistic estimate is decisive evidence against immediate multilevel implementation.

Two reductions are reported:

1. reduction relative to the currently realized field; and
2. incremental reduction relative to the best profiled single-level block shape.

The adoption gate uses the second and requires a worst-phase reduction of at least $2 \times$.

Evidence decision policy

The benchmark set must include at least one localized and one broad field. The outcome is:

```
insufficient_evidence
    if localized or broad coverage is missing

write_multilevel_specification
    if at least two fields are not adequately served by the completed
    single-level architecture and each has a phase-robust candidate with
    >= 2x incremental storage reduction within every scientific tolerance

retain_single_level
    otherwise
```

This decision is intentionally conservative. One promising field does not justify the new conservation and contouring machinery required by a production hierarchy.

Implemented benchmark matrix

The deterministic benchmark uses a 48^3 logical grid and covers:

1. a localized atomic cloud;
2. separated framework-vertex clouds;
3. overlapping oxygen clouds;
4. multimodal Na hopping occupancy;
5. projected framework-edge density;
6. atom-resolved framework-path density;
7. a smooth broad mobile-ion field.

The observed decision is `retain_single_level`.

Field	Production back-end	Active fraction	Current stored fraction	Best profiled single-level values	Best passing multilevel candidate
atomic localized	sparse	0.0136	0.1481	3,136	none
framework vertices separated	sparse	0.0321	0.4815	8,192	none
oxygen clouds overlapping	sparse	0.0300	0.1852	6,400	none
Na multimodal hopping	sparse	0.0438	0.1481	9,280	none
framework edges projected	sparse	0.1009	0.7037	23,360	none
framework atomic paths	sparse	0.1148	0.7037	22,016	$r = 4, q_f = 0.99, 2.90x$
mobile ion broad	dense	1.0000	1.0000	110,592	$r = 2, q_f = 0.90, 1.09x$

The projected-edge field is the only representative field not classified as adequately served by the current backend or an alternative single-level block size. It has no coarse/fine candidate that passes all scientific tolerances. The atom-resolved path field has a passing optimistic candidate, but the best 4^3 single-level block plan is already within the localized-storage policy anchor. The broad field remains properly served by dense storage and offers only 1.09x worst-phase incremental reduction.

The required two insufficient cases with at least 2x phase-robust incremental benefit are therefore absent.

Resource and failure policy

Profiling is bounded independently of production field preparation:

```
max_profile_nodes      4,000,000
max_phase_evaluations  2,000
max_workspace_bytes    512,000,000
```

The profiler fails before candidate evaluation when node or workspace limits are exceeded. It never alters a field, backend decision, cache, renderer, or scene plan.

Candidate factors incompatible with the logical shape are skipped and recorded by the absence of that factor. A benchmark with no localized or no broad field returns `insufficient_evidence` rather than a positive or negative architecture decision.

Determinism and serialization

Positive nodes are collected through the public node-access protocol and sorted by global C-order logical index. Coarse phases, factors, fine mass fractions, block shapes, and HDR fractions are evaluated in sorted deterministic order.

Every options, phase, candidate, block, field, and decision record is immutable and schema-versioned. Full field profiles round-trip through canonical JSON when phase records are included. Decision records always round-trip through canonical JSON.

Focused validation

LD6 focused tests certify:

- options and evidence-record JSON round trips;
- exact mass preservation;
- exact broad-uniform coarse reconstruction;
- deterministic repeated profiles;
- every periodic phase for factors two and four;
- factor skipping for incompatible logical shapes;
- bounded node-resource failure;
- localized/broad benchmark coverage requirements;
- `retain_single_level` and `write_multilevel_specification` policy branches;
- unchanged dense and sparse scientific production paths;
- unchanged LD4 selector and LD5 cache behavior.

Completion decision

LD6 completes the adaptive-density roadmap with the following normative conclusion:

Retain the single-level dense plus block-sparse architecture. Do not implement a general multilevel hierarchy at this time.

Future multilevel work may be reopened only with new representative evidence showing at least two production-relevant cases that fail the current resource policy and retain at least 2x phase-robust incremental benefit under the LD6 tolerances. Such work must begin with a new specification for conservative transfer, multilevel HDR integration, periodic coarse/fine ownership, and crack-free adaptive contouring.

Reference

1. Berger, M. J., and P. Colella. "Local Adaptive Mesh Refinement for Shock Hydrodynamics." *Journal of Computational Physics* **82** (1989): 64-84. DOI: 10.1016/0021-9991(89)90035-1.